

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10 - Formative Assessment 3

Formative Assessment (FA-3)

Class: Class 10

Assessment Type: Formative Assessment (FA-3)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Python Programming - Part 1

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

- a. Which of the following is the correct way to import a module in Python?
 - b. `import module`
 - c. `include module`
 - d. `require module`
 - e. `using module`
- b. What will be the output of `print(type([]))`?
 - c. `<class 'tuple'>`
 - d. `<class 'list'>`
 - e. `<class 'array'>`
 - f. `<class 'dict'>`
- c. Which method is used to read a line from a file in Python?
 - d. `readline()`
 - e. `getLine()`
 - f. `fetchLine()`
 - g. `line()`
- d. What does `with open('file.txt') as f:` do?
 - e. Opens file and closes it automatically after the block
 - f. Opens file and keeps it open forever
 - g. Creates a new file
 - h. Deletes the file
- e. Which exception is raised when dividing by zero in Python?
 - f. `ValueError`
 - g. `ZeroDivisionError`
 - h. `ArithmeticError`
 - i. `RuntimeError`

Section II: Short Answer (5 x 2 = 10 marks)

a. Write the output of the following code and explain:

```
data = [10, 20, 30, 40, 50]
result = list(filter(lambda x: x > 25, data))
print(result)
```

- b. Differentiate between `read()`, `readline()`, and `readlines()` file methods.
- c. Write a Python function that accepts a string and returns a dictionary with character frequency.
Example: "hello" → {'h': 1, 'e': 1, 'l': 2, 'o': 1}
- d. What is the purpose of try-except block? Write an example where it is used for file handling.
- e. Explain the difference between **syntax error** and **runtime error** with one example each.

Section III: Long Answer (5 x 1 = 5 marks)

- a. Write a Python program to manage student records using file handling:
- Store name, roll number, and marks of 5 students in a text file
 - Read the file and display all students sorted by marks (descending)
 - Find and display the student with highest marks
 - Count how many students scored above 75%

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Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Reads a text file and counts the number of words, lines, and characters
- Ignores extra spaces and counts unique words
- Displays the 5 most frequently used words
- Writes the summary to a new file

Part C: Notebook Maintenance (5 marks)

- a. Neatness and organisation of notebook (2 marks)
- b. Completion of all assigned classwork and homework (2 marks)
- c. Proper headings, dates, and indexing (1 mark)